

Tushar Chaugule

Pune, Maharashtra | +91-8767088815 | chauguletushar2021@gmail.com | [Portfolio](#) | [LinkedIn](#) | [GitHub](#)

Professional Summary

Computer Engineering undergraduate focused on cybersecurity and security engineering, with hands-on experience building endpoint security, IoT security, security monitoring, controlled response, local-machine security, and forensic workflows. Experienced with Python, Linux, networking, VAPT, web security, API security, security testing, and security architecture.

Technical Skills

Programming: C, C++, Python, JavaScript, TypeScript, SQL, Bash

Security: VAPT, Vulnerability Assessment, OWASP Top 10, Web Security, API Security, CVSS, CWE

Networking: Computer Networks, DNS, HTTP, TLS, Network Traffic Analysis, Security Headers, OSINT

Security Engineering: Security Monitoring, Threat Detection, Authentication, RBAC, Rate Limiting, Forensics, Audit Logging, Security Architecture, Security Testing

IoT/Embedded: IoT Security, Device Authentication, Capability Controls, Embedded Systems, ARM7TDMI Simulation, UART

Tools/Development: Burp Suite, Nmap, Gobuster, Wireshark, Maltego, Linux, FastAPI, React.js, Tauri, Node.js, Express.js, REST APIs, WebSocket, Git, GitHub, Pytest, Proteus

Cybersecurity Projects

- Rakshak — Local-First Cybersecurity Control Plane** *Python, FastAPI, Pydantic, React, TypeScript, Tauri* [\[Git-Hub\]](#)
 - Built a local-first endpoint security control plane for monitoring, deterministic security decisions, controlled response, investigation, forensics, and tamper-evident auditing.
 - Implemented a 35-capability security model (26 read-only, 9 mutating) with authority, policy, capability checks, human confirmation, and single-use time-limited authorization for privileged operations.
 - Implemented SHA-256 hash-chained JSONL auditing, correlated investigation workflows, security-boundary testing, and automated regression validation with 1,387 tests passing.
- Virtual IoT Security Laboratory — IoT Security Testbed** *Python, FastAPI, React 19, TypeScript, C, Proteus* [\[Git-Hub\]](#)
 - Built and deployed an end-to-end IoT security laboratory combining Proteus LPC2138 ARM7 simulation, virtual sensor/actuator nodes, telemetry ingestion, real-time monitoring, detection, response, and forensic investigation.
 - Implemented HMAC-based device authentication, capability/RBAC enforcement, rate limiting, device lifecycle controls, and six deterministic detections covering rogue devices, brute-force authentication, telemetry floods, sensor tampering, inactive devices, and impersonation.
 - Implemented SHA-256 hash-chained audit verification, automated quarantine, correlated forensic timelines, and seven reproducible attack scenarios; validated with 117/117 automated tests passing.
- KS Sentinel 2.0 — Secure Workspace OS** *React 18, Vite, Node.js, Express, MongoDB* [\[Git-Hub\]](#)
 - Designed a three-tier architecture separating the React Web OS, Secure Gateway, and Windows Local Agent for controlled local-machine operations without direct browser access.
 - Implemented capability-based filesystem operations, workspace isolation, path validation, agent lifecycle/heartbeat, restricted endpoints, and zero-shell system telemetry with automated security validation.

Leadership Experience

- Tech Team Lead** *Sep 2024 – Oct 2025*
RSCOE Mathematics Club, JSPM's RSCOE *Pune, Maharashtra*
 - Led technical execution, coordinated development tasks, and supported team members in implementing technical solutions.

Education

- B.Tech. in Computer Engineering** *2023 – 2027*
JSPM's Rajarshi Shahu College of Engineering *Pune, Maharashtra*
 - CGPA: 8.65/10** | **Minor Degree in Embedded Systems — ENTC Department:** Digital Design & Computer Organization; Microcontrollers & Laboratory; IoT; Embedded Systems & Laboratory.

Certifications & Training

CEH Theory/Practical Training — Completed | **NPTEL Ethical Hacking** — IIT Kharagpur
NPTEL Network Security — IIT Kharagpur | **AWS Cloud Practitioner** | **AWS Hands-on Workshop**